*2026 UROP IRT Program*

CALIT2

# Designing nanostructured high-entropy materials for neuromorphic computing memristors

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## Introduction

- After 50 years of success, the semiconductor industry is now battling the limits of **Moore's law**.
- Scientists have turned to **memristors** that couple storage and computation via non-von Neumann architectures to combat the latest demands of AI[1].
- This project investigates **InZnSnCuO$_x$** as a **resistive-switching (RS)** material to improve memristor switching performance.

## Background




**Fig 1.1 (a)** *RS memristor,* ***(b)*** *Vacancy ($V_o^{••}$) distribution of the RS memristor in a high-resistance state (HRS) and low-resistance state (LRS).*

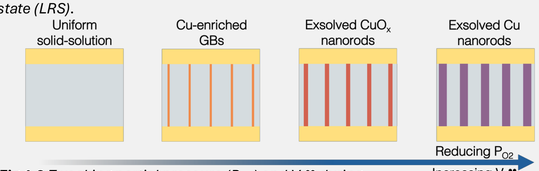
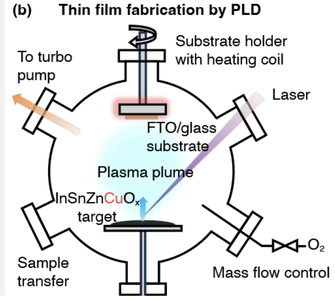



**Fig 1.2** *Tweaking partial pressure ($P_{O2}$) and $V_o^{••}$ during deposition can modify the nanophases observed.*

## Goal and Procedure

**Goal:** To enhance resistive switching performance by **embedding CuO$_x$ nanostructures** in oxide semiconductor thin films to regulate filament formation and improve switching stability.

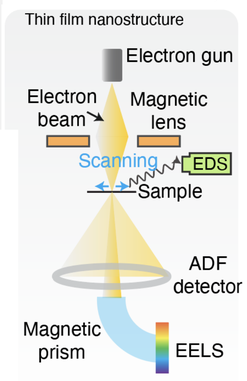
**(b) Thin film fabrication by PLD**




Parameters for thin film deposition

| Variant: Temperature during PLD (°C) | | | |
|---|---|---|---|
| InZnSnO$_x$ | 500 | 600 | 700 |
| InZnSnCuO$_x$ | 500 | 600 | 700 |

**(c) High-resolution structure characterization & data analysis**




by STEM imaging, EDS and EELS mapping

Nanostructure statistic analysis

by imaging processing

S/TEM sample preparation by SEM-FIB milling

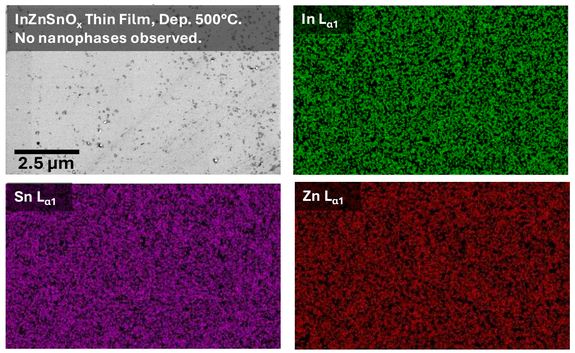
Nanostructure Characterization by SEM-EDS

Crystallinity by XRD

This project used facilities and instrumentation supported by the Bowman Lab at Calit2 and the IMRI.

## Results

### Copper nanophases observed in SEM/EDS of InZnSnCuO$_x$



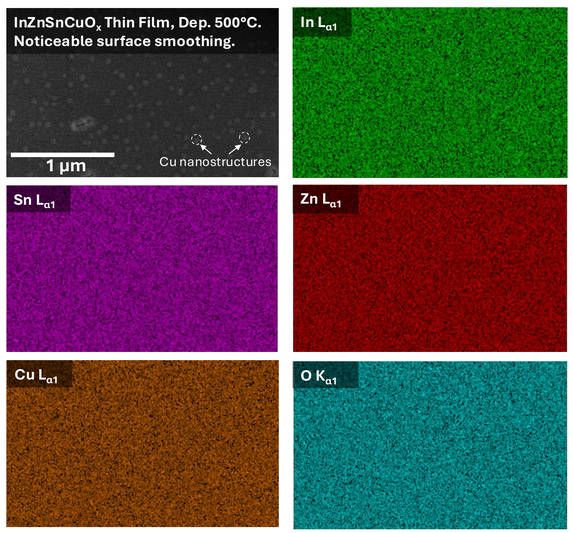




**Fig 3.2. (top)** InZnSnO$_x$ thin film deposited at 500°C compared to **(bottom)** 4 comp, room which enriched the grain boundaries with a copper nanophase.

### XRD Data to show InSnZnCuO$_x$ 4 component has copper oxide phase

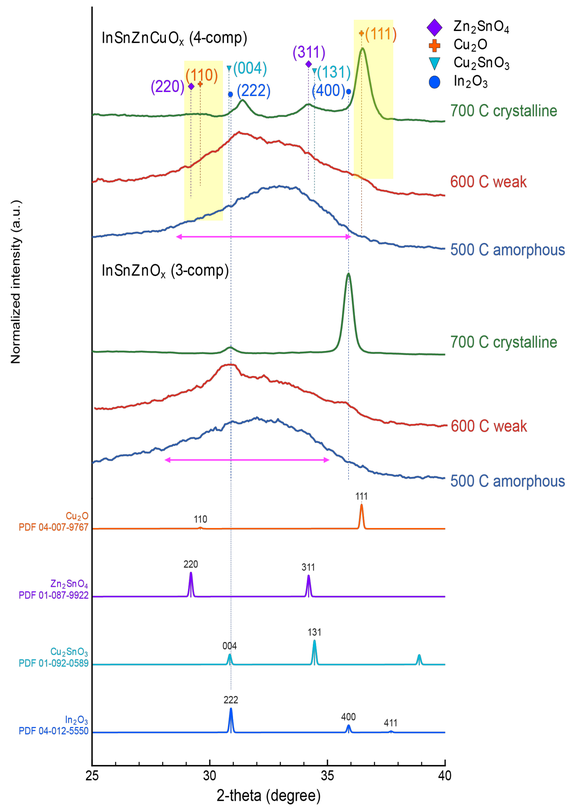



**Fig 4.1. (top)** XRD plot of InZnSnCuO$_x$ (4-comp); **(bottom)** of InZnSnO$_x$ (3-comp).

- All samples were crystallized at 600 °C;
- Compared with InSnZnO$_x$, the InSnZnCuO$_x$ film exhibits an additional secondary phase, attributed to the formation of a copper oxide phase and is consistent with the EDS results;
- The observed XRD peak shift suggests that Cu incorporation modifies the $In_2O_3$ lattice parameters.

### Tuning electronic conductivity with Cu-rich nanostructures

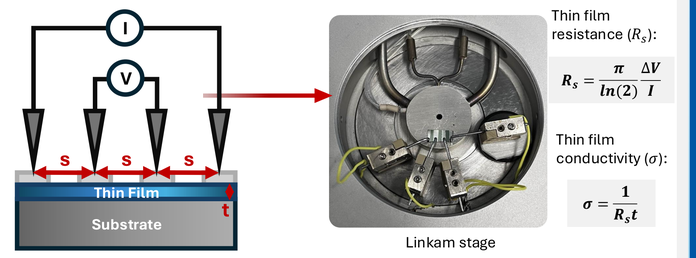



**Fig 5.1. (left)** Depiction of the 4-probe I-V test **(right)** Photograph of the experimental setup.

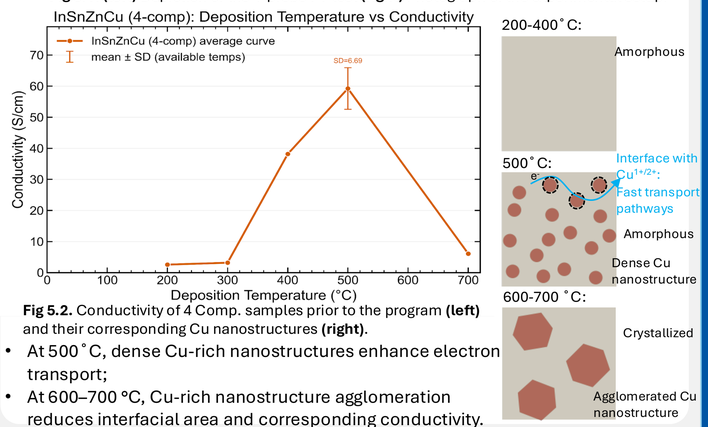



**Fig 5.2.** Conductivity of 4 Comp. samples prior to the program **(left)** and their corresponding Cu nanostructures **(right)**.

- At 500˚C, dense Cu-rich nanostructures enhance electron transport;
- At 600–700 °C, Cu-rich nanostructure agglomeration reduces interfacial area and corresponding conductivity.

## Conclusions

- PLD enables **diverse nanostructure design in oxide semiconductor films** for memristor applications through nonequilibrium synthesis and tunable processing parameters.
- Compared with InSnZnO$_x$, **InSnZnCuO$_x$** shows pronounced **Cu-rich/CuO-like nanostructure** formation at 500–700 °C, which may facilitate oxygen-vacancy redistribution and enhance resistive switching.
- The electronic conductivity of oxide semiconductor films can be tuned by controlling Cu-rich nanostructures.

## Current progress and Future Directions

| Completed tasks | | | In Progress (Current Status) | Future work | |
|---|---|---|---|---|---|
| Thin-Film **Fabrication** (PLD) | **Preliminary Nanostructure** Characterization (SEM-EDS) | **Crystallinity** Characterization (XRD) | **High-Resolution Nanostructure** Characterization (TEM imaging) | **Electrical Testing** (Probe Station) | Elucidating **Nanostructure–Property Relationships** |

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**Citations**

**[1]** Y. Xiao *et al.*, "A review of memristor: material and structure design, device performance, applications and prospects," *Sci Technol Adv Mater*, vol. 24, no. 1, p. 2162323, doi: 10.1080/14686996.2022.2162323.